

Laptop Request Process using ServiceNow - Capstone Project Documentation

- **1. Introduction**

Project Title:

Laptop Request Process using ServiceNow

- **2. Project Overview**

Purpose

To automate the complete laptop request process within an organization using the ServiceNow platform by replacing manual email-based requests with an automated approval and fulfillment workflow.

Features

- Laptop Request through Service Catalog
- Laptop Type Selection
- Business Justification
- Manager Approval Workflow
- Automatic Catalog Task Generation
- Email Notification
- Request Tracking
- Flow Designer Automation

Project Description

The Laptop Request Process is a ServiceNow-based automation solution developed to simplify and standardize the corporate laptop request process.

Traditionally, employees submit laptop requests through emails or manual forms, making the approval process slow and difficult to track. This project eliminates those manual activities by providing a self-service catalog where employees can request laptops online.

Employees access the Service Catalog, open the Laptop Request catalog item, select the required laptop type, enter the business justification, and submit the request.

Once submitted, ServiceNow automatically creates a Request (REQ) and a Requested Item (RITM). The Flow Designer workflow immediately starts and retrieves all submitted catalog variables.

The workflow automatically sends the request to the employee's manager for approval. The flow pauses until the manager either approves or rejects the request.

If the request is approved, the workflow automatically creates a Catalog Task (SCTASK) and assigns it to the Hardware Team for laptop provisioning.

After the Hardware Team completes the task, the workflow automatically sends an email notification informing the employee that the laptop is ready for pickup.

This solution improves transparency, reduces processing time, eliminates manual intervention, and provides complete visibility into the laptop request lifecycle.

- **3. Architecture**

Frontend

- Service Catalog
- Service Portal
- Catalog Variables

Backend

- Flow Designer
- Ask for Approval
- Create Catalog Task
- Send Email Action

Database

- sc_request
- sc_req_item
- sc_task

Architecture Description

The Laptop Request Process follows a layered architecture.

Presentation Layer

Employees access the Service Catalog through the Service Portal and submit laptop requests by selecting the laptop type and entering business justification.

Application Logic Layer

Flow Designer automates the complete workflow by retrieving catalog variables, requesting manager approval, creating fulfillment tasks, and sending email notifications.

Data Layer

The application stores request information in ServiceNow tables including:

- sc_request
- sc_req_item
- sc_task

These tables maintain request details, approval status, fulfillment information, and workflow history.

Notification Layer

After successful fulfillment, ServiceNow automatically sends an email notification to the requester informing them that the requested laptop is ready.

• 4. Setup Instructions

Prerequisites

- ServiceNow Developer Instance
- Admin Role
- Flow Designer Plugin Enabled

Step 1 : Create Catalog Item

Navigate to

Service Catalog → Maintain Items

Create a new catalog item named

Laptop Request

Fill in

Name: Laptop Request

Catalog: Service Catalog

Category: IT request

Short Description: Request a new laptop for official business use.

Save the record.

Step 2 : Add Variables

Create two variables.

Laptop Type

Type

Select Box

Choices

- Standard Window Laptops
- Developer MacBook Pro

Justification

Type

Multi Line Text

Save both variables.

Step 3 : Create Flow

Open

Flow Designer

Create a new Flow

Laptop Request Process Flow

Add Trigger

Service Catalog

Step 4 : Configure Flow

Add

Get Catalog Variables

Retrieve

- Laptop Type
 - Justification
-

Ask for Approval

Configure

Manager Approval

Approval Rule

Anyone Approves

If Approved

Add

If Condition

Approval State = Approved

Create Catalog Task

Assignment Group

Hardware

Short Description

Fulfill and provision requested laptop.

Send Email

Subject

Your Laptop Request is Complete!.

Body

Your laptop request has been provisioned. It is ready for pickup

Step 5 : Publish Flow

Save

Publish

Associate the Flow with the Catalog Item.

Step 6 : Testing

Open

Laptop Request

Click

Try It

Submit a request

Approve the request

Verify

- Catalog Task Creation
 - Email Notification
 - RITM Status
-

- **5. Folder / Component Structure**

Service Catalog Layer

Contains

- Laptop Request Catalog Item
 - Catalog Variables
 - Service Portal Form
-

Workflow Layer

Contains

- Flow Designer
 - Approval Action
 - If Logic
 - Catalog Task
 - Email Notification
-

Data Layer

Uses

- sc_request
- sc_req_item
- sc_task

Stores

- Request Details
 - Approval Status
 - Catalog Tasks
-

Notification Layer

Uses

- ServiceNow Email Notification

Provides

- Completion Email
 - Request Updates
-

- **6. Running the Application**

Step 1

Open Service Portal

Step 2

Open

Laptop Request

Step 3

Select

Laptop Type

Step 4

Enter

Justification

Step 5

Click

Order Now

Step 6

Flow Designer automatically

- Retrieves Variables
 - Sends Approval
-

Step 7

Manager approves request.

Step 8

Flow creates

Catalog Task

Step 9

Hardware Team provisions laptop.

Step 10

Email notification is automatically sent.

- **7. Automation Logic**

The Laptop Request Process is implemented entirely using **Flow Designer**.

The workflow performs the following automation:

- Retrieve Catalog Variables
- Manager Approval
- Approval Decision
- Create Catalog Task
- Send Completion Email

The workflow is automatically triggered whenever the Laptop Request catalog item is submitted.

- **8. Authentication**

The application uses ServiceNow Role-Based Access Control.

User

Can

- Submit Laptop Request
- View Own Requests

Manager

Can

- Approve
- Reject Requests

Admin

Can

- Configure Catalog Items
- Modify Flows
- Monitor Requests

Security is implemented using

- Login Authentication
 - ACLs
 - HTTPS
-

- **9. User Interface**

The application provides a simple Service Catalog interface.

Users can

- Select Laptop Type
- Enter Justification
- Submit Request

After submission

Users can

- Track Request
- View Approval Status
- Receive Email Notification

The interface is designed to be

- Simple
- Responsive
- Easy to Use

- **10. Testing**

Test Case 1

Laptop Request Submission

Expected

Request created successfully.

Result

Pass

Test Case 2

Mandatory Variable Validation

Expected

Request cannot be submitted without justification.

Result

Pass

Test Case 3

Manager Approval

Expected

Approval request generated.

Result

Pass

Test Case 4

Catalog Task Creation

Expected

Task created after approval.

Result

Pass

Test Case 5

Email Notification

Expected

Requester receives completion email.

Result

Pass

- **11. Known Issues**

- Email delivery depends on SMTP configuration.
 - Approval process waits until the manager responds.
 - Hardware assignment group must exist before task creation.
 - Internet connection is required to access ServiceNow.
 - Notification delays may occur during maintenance windows.
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- **12. Future Enhancements**
- Integration with Asset Management (ITAM)
- Automatic Laptop Inventory Check
- Procurement System Integration
- Mobile Approval Support
- Multi-Level Approval Workflow
- SLA Monitoring
- Dashboard & Analytics
- AI-Based Laptop Recommendation
- Microsoft Teams & Slack Notifications
- Automated Asset Allocation